

INST327 – Assignment 3

Questions

Use the dataset given below to answer both questions (Top-Down Design and Normalization). Normalization in Question 2 should be completed through 1NF, 2NF, and 3NF. Show the normalization process in a file (e.g., creating tables and drawing relationship lines in MS Excel, Google Sheets, or some other tool).

Submit your files through the link available on Canvas. See the instructions for file name formats.

The presentation format of your normalization process should mimic the normalization class exercises and the examples provided in Canvas.

To complete this assignment successfully, ensure you address all of the points included in each question.

Q. 1) Top-Down Design (40 points)

- Show your work in the Top-Down Design steps, as discussed in Week 4:
 - a. Identify data elements
 - b. Sub-divide each element into its smallest useful components
 - c. Identify tables (entities) and assign columns (attributes)
 - d. Identify relationships between tables (primary and foreign keys)

Q. 2) Normalization (60 points)

- **DO NOT** use the results of the top-down design as input or target for the normalization question! **EXCEPTION:** You may use the data elements list as input or you may choose to re-create the data elements list for normalization.
- Normalize the flat dataset through each of the 1NF, 2NF, and 3NF steps **separately, explicitly, and in sequence.**
- Do not skip any normalization steps.
- In each normalization step, you should only resolve the issue(s) that pertain to the given step. For example, do not resolve transitive dependencies in 1NF or 2NF, as transitive dependencies are 3NF business.
- In each step, list all of the tables, including those that do not change in the given step. Identify and show primary keys and foreign keys in each step. Identify composite primary keys where they exist.
- **For each step, describe the changes you made in a sentence or two.** This does not need to be long but should outline the changes you made and why you did them in the step. This accounts for 5 points for each step (1NF, 2NF, and 3NF).
- Show the relationships between the tables in each form by drawing lines/arrows between corresponding PKs and FKs.
- The sample records are provided for your reference to help you understand the nature of each field/attribute. You should not include them in your work.

Guitars Dataset

- Things to note for this dataset:
 - The sample data is provided to help you understand what kind of information is in each column, but you should not include the data in your answers.
 - The sample demonstrates information about a few guitars. However, there may be more guitars, manufacturers, and genres.
 - The sample data shows that manufacturers can produce multiple kinds of guitars but each guitar is produced by only one manufacturer.
 - The sample data shows that one kind of guitar can be used in multiple genres and genres can use multiple kinds of guitars.

Guitar ID	Guitar Name	Guitar Price	Manufacturer Name	Manufacturer Location	Genre Name 1	Genre Tempo 1	Genre Name 2	Genre Tempo 2
1	Stratocaster	750	Fender	Corona, California	Rock	Fast	Blues	Slow
2	Telecaster	800	Fender	Corona, California	Country	Moderate	Blues	Slow
3	Les Paul	3000	Gibson	Nashville, Tennessee	Rock	Fast		
4	SG	2000	Gibson	Nashville, Tennessee	Metal	Very Fast	Rock	Fast